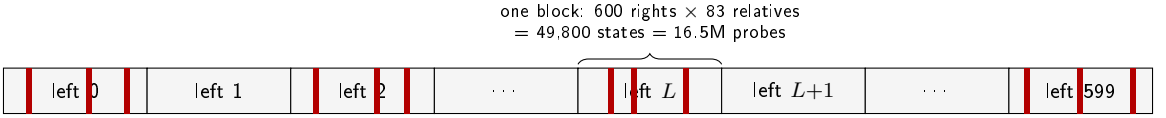
(a) The state vector: 29,880,000 states, left-orbit-major



red: the 12 hit states of one representation $Q = \pm A \pm B \pm C \pm D$, three in each of the blocks of A, B, C, D;
blocks whose orbits are in few representations are hit-free for millions of probes

(b) Baseline: start = hash$(Q) \bmod S$, then consecutive states (wraps)



the probe count to the first hit depends on which blocks the start lands in: a start late in a sparse block runs into the next one;
archived tails: 43,541, 54,512 and 53,114 consecutive hit-free states (w00, w05, w13; law mean 5,204)

(c) Fix: the same states, each once, in a keyed Feistel order (key = the same hash)



consecutive states fall in unrelated blocks, so the hit times are close to independent and the probe count to the first hit is close to geometric
with mean $r/(A \cdot \#\text{states}) = 1.73$M; per-state scan, canonicalizer, relation check and recovery are unchanged; exactness: every state is still reachable